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Reticle pod inner pod having dissimilar material at contact surface interfaces

Abstract

Reticle pod inner pods include a cover and a baseplate, with the cover and the baseplate contacting one another at contact surfaces. The contact surfaces of the cover and the baseplate each include different materials. The different materials can each be metals. The different materials can differ in hardness. The difference in hardness can be 50 Brinell hardness or greater. One of the different materials can be a ductile material, having elongation at break of 25% or greater. Methods can include providing the cover and the baseplate each including a first material, and providing a second, different material at the contact surfaces of one of the cover or the baseplate.

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Background/Summary

PRIORITY (1) This disclosure claims priority to U.S. provisional patent No. 63/292,402 with a filing date of Dec. 21, 2021, which is incorporated by reference herein.

FIELD
(1) This disclosure is directed to reticle pods having dissimilar materials at interfacing contact surfaces.
BACKGROUND
(2) Reticle pods can be used to store and convey reticles, for example reticles subjected to photolithography such as extreme ultraviolet (EUV) photolithography. The reticle pods can include outer pods and inner pods. The inner pods can include metals, typically hard metal materials. The inner pods typically include a baseplate and a cover which contact one another when the inner pod is closed, for example when a reticle is contained within.

SUMMARY

(3) This disclosure is directed to reticle pods having dissimilar materials at interfacing contact surfaces.

(4) Contact surfaces of covers and baseplates of reticle pod inner pods can use different materials at the respective contact surfaces. The different materials can differ in hardness to provide a relatively soft and/or ductile material at one of the contact surfaces and a relatively harder material at the other. The difference in hardness between the different materials can be selected to reduce particle generation due to wear and/or the incidence of galling at the contact surfaces. The reduction in particle generation provided by the use of the dissimilar materials can improve yields and reduce damage to or loss of reticles in storage, transportation, and photolithography processes using inner pods according to embodiments.

(5) In an embodiment, a reticle pod includes a baseplate and a cover. The baseplate and the cover configured to accommodate a reticle. The baseplate includes one or more first contact surfaces on a side of the baseplate configured to face the cover. The cover includes one or more second contact surfaces, each of the one or more second contact surfaces configured to contact at least one of the one or more first contact surfaces. The one or more first contact surfaces are formed of a first material. The one or more second contact surfaces are formed of a second material. The second material is different from the first material.

(6) In an embodiment, when the one or more first contact surfaces contact the one or more second contact surfaces, a seal is formed at the interface of the one or more first contact surfaces with the one or more second contact surfaces.

(7) In an embodiment, the first material and the second material differ in hardness by at least 50 Brinell hardness. In an embodiment, one of the first material and the second material is gold, and the other of the first material and the second material is selected from the group consisting of nickel and chrome. In an embodiment, at least one of the first material and the second material are metals. In an embodiment, one of the first material and the second material is provided as a coating on a respective one of the baseplate or the cover. In an embodiment, at least one of the first material or the second material has a ductility value D of 0.62 or greater. In an embodiment, at least one of the first material or the second material is non-oxidizing.

(8) In an embodiment, a method of manufacturing a reticle pod includes forming a baseplate of a first material, where the baseplate includes one or more first contact regions. The method further includes forming a cover of the first material, the cover including one or more second contact regions, where the one or more second contact regions are configured to oppose the one or more first contact regions when the baseplate and the cover are joined. The method also includes applying a second material, different from the first material, to one of the cover or the baseplate at least at the one or more first contact regions or the one or more second contact regions to form a contact surface.

(9) In an embodiment, the first material and the second material differ in hardness by at least 50 Brinell hardness. In an embodiment, the first material is selected from the group consisting of nickel and chrome, and the second material is gold. In an embodiment, each of the first material and the second material are metals. In an embodiment, applying the second material comprises coating the either of the one or more first contact regions or the one or more second contact regions with the second material. In an embodiment, the second material has ductility value D of 0.62 or greater. In an embodiment, the second material is non-oxidizing.

(10) In an embodiment, the method further includes applying a third material, different from the second material, to the other of the cover or the baseplate at least at the one or more first contact regions or the one or more second contact regions to form a second contact surface. In an embodiment, the third material and the second material differ in hardness by at least 50 Brinell hardness. In an embodiment, the third material is selected from the group consisting of nickel and chrome, and the second material is gold. In an embodiment, the third material and the second

material are metals. In an embodiment, the third material is non-oxidizing.

(11) In an embodiment, a method of storing a reticle includes placing the reticle into a reticle pod including a cover and a baseplate. The method further includes bringing one or more first contact surfaces provided on the cover into contact with one or more second contact surfaces provided on the baseplate. The one or more first contact surfaces are formed of a first material and the one or more second contact surfaces are formed of a second material. The second material is different from the first material. In an embodiment, the first material and the second material differ in hardness by at least 50 Brinell hardness. In an embodiment, at least one of the first material and the second material has ductility value D of 0.62 or greater. In an embodiment, at least one of the first material and the second material is non-oxidizing.

Description

DRAWINGS

(1) FIG. **1** shows a reticle pod according to an embodiment.

(2) FIG. **2** shows a sectional view of contact surfaces of a reticle pod inner pod according to an embodiment.

(3) FIG. **3** shows a sectional view of contact surfaces of a reticle pod inner pod according to an embodiment.

(4) FIG. **4** shows a view of contact surfaces of a reticle pod inner pod according to an embodiment.

(5) FIG. **5** shows a sectional view of contact surfaces of a reticle pod inner pod according to an embodiment.

DETAILED DESCRIPTION

(6) This disclosure is directed to reticle pods having dissimilar materials at interfacing contact surfaces.

(7) FIG. **1** shows a reticle pod according to an embodiment. Reticle pod **100** includes an outer pod **102** including pod dome **104** and pod door **106**. Reticle pod **100** further includes inner pod **108**, which includes cover **110** and baseplate **112**. Cover **110** includes one or more cover contact surfaces **114**. Baseplate **112** includes one or more baseplate contact surfaces **116**. Reticle **118** can be contained within the reticle pod **100**.

(8) Outer pod **102** forms an exterior of reticle pod **100** when reticle pod **100** is fully assembled. Outer pod **102** is configured to form an internal space capable of accommodating inner pod **108**. Outer pod **102** can be formed of pod dome **104** and pod door **106**. Pod dome **104** and pod door **106** can be configured to be joined to one another, for example using a latching mechanism (not shown). The pod dome **104** and pod door **106** can be configured such that the inner pod **108** is retained within the internal space defined by the pod dome **104** and pod door **106**.

(9) Inner pod **108** is a pod configured to accommodate the reticle **118**. Inner pod **108** includes the cover **110** and the baseplate **112**. When inner pod **108** is assembled, the cover **110** and the baseplate **112** contact one another at the respective cover contact surfaces **114** and baseplate contact surfaces **116**. The inner pod **108** is configured such that it can fit within the internal space of outer pod **102**.

(10) Cover **110** forms a portion of inner pod **108**. Cover **110** can be formed of any suitable material, for example a metal material such as aluminum. Cover **110** can include a coating or plating on the base material. The cover **110** includes cover contact surfaces **114**. The cover contact surfaces **114** are disposed on a side of cover **110** that faces baseplate **112** when the inner pod **108** is assembled. In an embodiment, the cover contact surfaces **114** are provided as a single continuous surface. In an embodiment, the cover contact surfaces **114** are formed of the base material of the cover **110**. In an embodiment, the cover contact surfaces **114** are plated and/or coated with a material. In an embodiment, the cover contact surfaces **114** are a plurality of discrete contact surfaces. In an embodiment, cover contact surfaces **114** are provided on a plurality of contact strips, such as

contact strips **308** shown in FIG. **3** and described below. In an embodiment, cover contact surfaces **114** are provided on a plurality of contact points such as contact strips **408** shown in FIG. **4** and described below. In an embodiment, cover contact surfaces **114** are provided on one or more contact inserts such as contact inserts **508** as shown in FIG. **5** and described below.

(11) Baseplate **112** forms another portion of inner pod **108**. Baseplate **112** can be formed of any suitable material, for example a metal material such as aluminum. Baseplate **112** can include a coating or plating on the base material. The baseplate **112** includes baseplate contact surfaces **116**. In an embodiment, the baseplate contact surfaces **116** are formed of the base material of the baseplate **112**. In an embodiment, the baseplate contact surfaces **116** are plated and/or coated with a material. The baseplate contact surfaces **116** are disposed on a side of baseplate **112** that faces baseplate **112** when the inner pod **108** is assembled. In an embodiment, the baseplate contact surfaces **116** are provided as a single continuous surface. In an embodiment, the baseplate contact surfaces **116** are a plurality of discrete contact surfaces. In an embodiment, baseplate contact surfaces **116** are provided on a plurality of contact strips, such as contact strips **308** shown in FIG. **3** and described below. In an embodiment, baseplate contact surfaces **116** are provided on a plurality of contact points such as contact strips **408** shown in FIG. **4** and described below. In an embodiment, baseplate contact surfaces **116** are provided on one or more contact inserts such as contact inserts **508** as shown in FIG. **5** and described below. In an embodiment, the cover contact surfaces **114** and the baseplate contact surfaces **116** are configured such that a seal is formed when reticle pod **100** is assembled. The seal can be sufficient to maintain a pressure difference between an interior of the reticle pod inner pod **108** and an exterior of inner pod **108** under conditions such as when a vacuum is drawn external to inner pod **108** or the like. In an embodiment, the seal is formed by direct contact or close proximity between the cover contact surfaces **114** and the baseplate contact surfaces **116**, as possible based on manufacturing tolerances, wear, or other such variances.

(12) The cover contact surfaces **114** and baseplate contact surfaces **116** are respectively formed such that dissimilar materials are in contact at the interface between the cover contact surfaces **114** and baseplate contact surfaces **116**. The dissimilar materials can be any two different materials that are different from one another. In an embodiment, the dissimilar materials can be selected to include a relatively hard material and a relatively soft material. Non-limiting examples of the relatively hard material include nickel, chrome, diamond-like coatings (DLC), chromium carbon nitride, and/or other materials of similar hardness. Non-limiting examples of the relatively soft material include gold, platinum, and other materials of similar hardness. In an embodiment, the dissimilar materials are selected to have a difference in hardness sufficient to avoid galling, for example by having a difference in hardness of at least 50 Brinell hardness (HB). In an embodiment, at least one of the dissimilar materials is a ductile material, having a physical ductility value D of 0.62 or greater. Non-limiting examples of ductile materials include aluminum, copper, tin, silver, platinum, niobium, lead, and gold. In embodiments, ductility can be characterized by the elongation at break of the material. In an embodiment, both of cover contact surface **114** and baseplate contact surface **116** include a ductile material. In an embodiment, only one of cover contact surface **114** and baseplate contact surface **116** includes the ductile material. In an embodiment, one or both of the materials of cover contact surface **114** and baseplate contact surface **116** is a metal. In an embodiment, at least one of the materials of the cover contact surface **114** or baseplate contact surface **116** is a polymer or composite material. A non-limiting example of a polymer material is a polyether-ether-ketone (PEEK) material, which can be included in a composite material such as a filled PEEK material. In embodiments, the materials selected to be the dissimilar materials at cover contact surface **114** and baseplate contact surface **116** can be selected based on wear properties of the materials such as particle generation, visible wear over use cycles, or the like. The wear properties can be determined for the materials and used in their selection individually or in combination with one another.

(13) FIG. **2** shows a sectional view of contact surfaces of a reticle pod inner pod according to an embodiment. Reticle pod inner pod **200** includes cover **202** and baseplate **204**. Cover **202** includes cover contact surface **206**. Baseplate **204** includes baseplate contact surface **210**. The cover contact surface **206** and the baseplate contact surface **208** are configured to contact one another when reticle pod inner pod **200** is assembled.
(14) Cover **202** forms a portion of reticle pod inner pod **200**. Cover contact surface **206** is provided where cover **202** is configured to contact baseplate **204** when reticle pod inner pod **200** is assembled. In the embodiment shown in FIG. **2**, the cover contact surface **206** provides the base material of cover **202** where cover **202** contacts baseplate **204**. In embodiments, cover contact surface **206** can instead be coated or plated as shown for baseplate contact surface **208** to provide the material at the interface with baseplate contact surface **208**.
(15) Baseplate **204** forms another portion of reticle inner pod **200**. Baseplate contact surface **208** is provided where baseplate **204** contacts cover **202** when reticle inner pod **200** is assembled. In the embodiment shown in FIG. **2**, the baseplate contact surface **208** is coated or plated with a material. The coating at baseplate contact surface **208** is a dissimilar materials form that provided at cover contact surface **206**. In other embodiments, cover contact surface **206** can be coated or plated with a material, in addition to or in place of the coating provided on baseplate contact surface **208** as shown in FIG. **2**, so long as the material at cover contact surface **206** and baseplate contact surface **208** are dissimilar to one another. The dissimilar materials can differ in ductility, hardness, or the like as discussed above.
(16) Cover contact surface **206** and baseplate contact surface **208** can be configured to provide dissimilar materials where the cover and baseplate contact surfaces **206**, **208** meet one another. In an embodiment, one of the dissimilar materials can be a base material of one of the cover or the baseplate. In an embodiment, at least one of the dissimilar materials can be a coating or plating applied to cover **202** or baseplate **204**. In an embodiment, both cover **202** and baseplate **204** can each have one of the dissimilar materials applied as a coating or plating, as shown for baseplate contact surface **208** in FIG. **2**. In an embodiment, one of cover contact surface **206** or baseplate contact surface **208** is a coating or plating applied to cover **202** or baseplate **204**, and the other of cover contact surface **206** or baseplate contact surface **208** is provided as contact strips such as contact strips **308** shown in FIG. **3** and described below. In an embodiment, one of cover contact surface **206** or baseplate contact surface **208** is a coating or plating applied to cover **202** or baseplate **204**, and the other of cover contact surface **206** or baseplate contact surface **208** is provided as contact strips such as contact points **408** shown in FIG. **4** and described below. In an embodiment, one of cover contact surface **206** or baseplate contact surface **208** is a coating or plating applied to cover **202** or baseplate **204**, and the other of cover contact surface **206** or baseplate contact surface **208** is provided as contact strips such as contact inserts **508** shown in FIG. **5** and described below.
(17) FIG. **3** shows a sectional view of contact surfaces of a reticle pod inner pod according to an embodiment. Reticle pod inner pod **300** includes cover **302** and baseplate **304**. Cover **302** includes cover contact surface **306**. Baseplate **304** includes a plurality of contact strips **308** formed on baseplate **304**.
(18) Cover **302** forms a portion of reticle pod inner pod **300**. Cover contact surface **306** is provided on cover **302** where cover **302** is configured to contact baseplate **304**, for example at locations corresponding to contact strips **308** when cover **302** and baseplate **304** are combined to form reticle pod inner pod **300**. In an embodiment, cover contact surface **306** is a base material. In an embodiment, cover contact surface **306** includes a coating or plating as the material presented at the cover contact surface **306**.
(19) Baseplate **304** forms another portion of reticle pod inner pod **300**. Contact strips **308** are strips of additional material formed on the baseplate **304**. The additional material is a material dissimilar to the material at cover contact surface **306**. The contact strips **308** can be directly applied to

baseplate **304**, for example depositing the contact strips **308**. In an embodiment, the contact strips **308** can be attached through any suitable attachment, such as adhesives or the like.

(20) While FIG. **3** shows the contact strips **308** as being formed on baseplate **304**, the contact strips **308** can instead be formed on the cover **302**, and baseplate **304** can instead provide contact surfaces of base material of the baseplate **304**, a coating provided on the baseplate **304**, or the baseplate can include contact points such as contact points **408** as discussed below and shown in FIG. **4**, or one or more contact inserts such as contact insert **508** as discussed below and shown in FIG. **5**.

(21) FIG. **4** shows a sectional view of contact surfaces of a reticle pod inner pod according to an embodiment. Reticle pod inner pod **400** includes cover **402** and baseplate **404**. Cover **402** includes cover contact surface **406**. Baseplate **404** includes a plurality of contact points **408** formed on baseplate **404**.

(22) Cover **402** forms a portion of reticle pod inner pod **400**. Cover contact surface **406** is provided on cover **402** where cover **402** is configured to contact baseplate **404**, for example at locations corresponding to contact points **408** when cover **402** and baseplate **404** are combined to form reticle pod inner pod **400**. In an embodiment, cover contact surface **406** is a base material. In an embodiment, cover contact surface **406** includes a coating or plating as the material presented at the cover contact surface **406**.

(23) Baseplate **404** forms another portion of reticle pod inner pod **400**. Contact points **408** are provided on baseplate **404**. Contact points **408** are arranged such that the cover contact surface **406** contacts the baseplate **404** at the contact points **408** when baseplate **404** and cover **402** are joined together to form reticle pod inner pod **400**. Contact points **408** can be a series of discrete formed on the baseplate **404**. Contact points **408** can be shapes such as dots, squares, or the like. Contact points **408** can be arranged in an array on the baseplate **404**. Contact points **408** are formed of a material dissimilar to the material at cover contact surface **406**.

(24) While FIG. **4** shows the contact points **408** as being formed on baseplate **404**, contact points **408** can instead be formed on cover **402**, with baseplate **404** providing contact surfaces of bare material or a plating or coating as for baseplate contact surfaces **208** as shown in FIG. **2** and discussed above, contact strips **308** as shown in FIG. **3** and discussed above, and/or contact inserts **508** as described below and shown in FIG. **5**. The contact points **408** and the material of cover **402** or baseplate **404** contacted by said contact points **408** are dissimilar materials.

(25) FIG. **5** shows a view of contact surfaces of a reticle pod inner pod according to an embodiment. Reticle pod inner pod **500** includes cover **502** and baseplate **504**. Cover **502** includes cover contact surface **506**. Baseplate **504** includes a plurality of contact inserts **508** provided on baseplate **504**. Contact inserts **510** can be joined to the baseplate **504** using channels **510**.

(26) Cover **502** forms a portion of reticle pod inner pod **500**. Cover contact surface **506** is provided on cover **502** where the cover **502** is configured to contact baseplate **504** when reticle pod inner pod **500** is assembled. The cover contact surface **506** can be a base material of the cover **502**, a coating or plating, or any other suitable contact surface, for example, contact strips such as contact strips **308** as shown in FIG. **3** and described above or contact points **408** as shown in FIG. **4** and described above.

(27) Baseplate **504** forms another portion of reticle pod inner pod **500**. Baseplate **504** includes contact inserts **508** provided on baseplate **504**. The contact inserts **508** are fixed to baseplate **504**, for example, through mechanical engagement such as interfacing features or a press fit between portions of the contact inserts **504** and portions of the baseplate. In an embodiment, channels **510** can receive a portion of contact inserts **508** to retain the contact inserts **508** in place. The channels **510** can be depressions or grooves formed in a surface of baseplate **504**. In an embodiment, the channels **510** can be through holes formed in the baseplate **504**. In an embodiment, channels **510** are sized such that retention protrusions **512** formed on contact inserts **508** opposite the side configured to contact cover contact surface **506** are press-fit within channels **510** to retain contact inserts **508** in place on baseplate **504**. In an embodiment, the retention protrusions **512** can be fixed

to the channels **510** using an adhesive. In an embodiment, the retention protrusions **512** include retention features that engage with the baseplate **504** to retain the contact inserts **508** in place on baseplate **504**. Cover contact surfaces **506** and contact inserts **508** are made of materials selected such that the contacting portions of cover contact surfaces **506** and contact inserts **508** respectively are formed of dissimilar materials to one another.

(28) While FIG. **5** shows the contact inserts **508** and channels **510** being provided on baseplate **504**, in embodiments, the channels **510** can be formed in cover **502** and contact inserts **508** provided on cover **502**. In these embodiments, baseplate **504** can instead provide contact surfaces of base material of the baseplate **504**, a coating provided on the baseplate **504**, or the baseplate **504** can include contact strips such as contact points **308** as discussed above and shown in FIG. **3**, or one or contact points such as contact points **408** as discussed above and shown in FIG. **4**.

(29) Aspects:

(30) It is understood that any of aspects 1-8 can be combined with any of aspects 9-20.

(31) Aspect 1. A reticle pod, comprising: a baseplate and a cover, the baseplate and the cover configured to accommodate a reticle, wherein: the baseplate includes one or more first contact surfaces on a side of the baseplate configured to face the cover; the cover includes one or more second contact surfaces, each of the one or more second contact surfaces configured to contact at least one of the one or more first contact surfaces; the one or more first contact surfaces are formed of a first material; the one or more second contact surfaces are formed of a second material, the second material being different from the first material.

(32) Aspect 2. The reticle pod according to aspect 1, wherein when the one or more first contact surfaces contact the one or more second contact surfaces, a seal is formed at the interface of the one or more first contact surfaces with the one or more second contact surfaces.

(33) Aspect 3. The reticle pod according to any of aspects 1-2, wherein the first material and the second material differ in hardness by at least 50 Brinell hardness.

(34) Aspect 4. The reticle pod according to any of aspects 1-3, wherein one of the first material and the second material is gold, and the other of the first material and the second material is selected from the group consisting of nickel and chrome.

(35) Aspect 5. The reticle pod according to any of aspects 1-4, wherein each of the first material and the second material are metals.

(36) Aspect 6. The reticle pod according to any of aspects 1-5, wherein one of the first material and the second material is provided as a coating on a respective one of the baseplate or the cover.

(37) Aspect 7. The reticle pod according to any of aspects 1-6, wherein at least one of the first material or the second material has a ductility value D of 0.62 or greater.

(38) Aspect 8. The reticle pod according to any of aspects 1-7, wherein at least one of the first material or the second material is non-oxidizing.

(39) Aspect 9. A method of manufacturing a reticle pod, comprising: forming a baseplate of a first material, the baseplate including one or more first contact regions; forming a cover of the first material, the cover including one or more second contact regions, the one or more second contact regions configured to oppose the one or more first contact regions when the baseplate and the cover are joined; and applying a second material, different from the first material, to one of the cover or the baseplate at least at the one or more first contact regions or the one or more second contact regions to form a contact surface.

(40) Aspect 10. The method according to aspect 9, wherein the first material and the second material differ in hardness by at least 50 Brinell hardness.

(41) Aspect 11. The method according to any of aspects 9-10, wherein the first material is selected from the group consisting of nickel and chrome, and the second material is gold.

(42) Aspect 12. The method according to any of aspects 9-11, wherein each of the first material and the second material are metals.

(43) Aspect 13. The method according to any of aspects 9-12, wherein applying the second material

comprises coating the either of the one or more first contact regions or the one or more second contact regions with the second material.
(44) Aspect 14. The method according to any of aspects 9-13, wherein the second material has ductility value D of 0.62 or greater.
(45) Aspect 15. The method according to any of aspects 9-14, wherein the second material is non-oxidizing.
(46) Aspect 16. The method according to any of aspects 9-15, further comprising applying a third material, different from the second material, to the other of the cover or the baseplate at least at the one or more first contact regions or the one or more second contact regions to form a second contact surface.
(47) Aspect 17. A method of storing a reticle, comprising: placing the reticle into a reticle pod including a cover and a baseplate, and bringing one or more first contact surfaces provided on the cover into contact with one or more second contact surfaces provided on the baseplate, wherein the one or more first contact surfaces are formed of a first material and the one or more second contact surfaces are formed of a second material, the second material being different from the first material.
(48) Aspect 18. The method according to aspect 17, wherein the first material and the second material differ in hardness by at least 50 Brinell hardness.
(49) Aspect 19. The method according to any of aspects 17-18, wherein at least one of the first material and the second material has ductility value D of 0.62 or greater.
(50) Aspect 20. The method according to any of aspects 17-19, wherein at least one of the first material and the second material is non-oxidizing.
(51) The examples disclosed in this application are to be considered in all respects as illustrative and not limitative. The scope of the invention is indicated by the appended claims rather than by the foregoing description; and all changes which come within the meaning and range of equivalency of the claims are intended to be embraced therein.

Claims

1. A container, comprising: a reticle pod including: a baseplate and a cover, the baseplate and the cover configured to accommodate a reticle, wherein: the baseplate includes one or more first contact surfaces on a side of the baseplate configured to face the cover; the cover includes one or more second contact surfaces, each of the one or more second contact surfaces configured to contact at least one of the one or more first contact surfaces; the one or more first contact surfaces are formed of a first material; the one or more second contact surfaces are formed of a second material, the second material being different from the first material, wherein one of the first material and the second material is provided as a coating on a respective one of the baseplate or the cover and wherein each of the first material and the second material are metals.
2. The reticle pod of claim 1, wherein when the one or more first contact surfaces contact the one or more second contact surfaces, a seal is formed at the interface of the one or more first contact surfaces with the one or more second contact surfaces.
3. The reticle pod of claim 1, wherein the first material and the second material differ in hardness by at least 50 Brinell hardness.
4. The reticle pod of claim 1, wherein one of the first material and the second material is gold, and the other of the first material and the second material is selected from the group consisting of nickel and chrome.
5. The reticle pod of claim 1, wherein at least one of the first material or the second material has a ductility value D of 0.62 or greater.

6. The reticle pod of claim 1, wherein at least one of the first material or the second material is non-oxidizing.